

1. EX. 2.1.2.620, Saver 3

b)

$$x + 2y - z = 2 \quad (1)$$

$$3y + x = 4 \quad (2)$$

$$2x - y + z = 2 \quad (3)$$

$$\Rightarrow 2x - y + z - 2(x + 2y - z) = 2 - 2(2)$$

$$\Rightarrow 2x - y + z - 2x - 4y + 2z = -2$$

$$\Rightarrow -5y + 3z = -2 \quad (3')$$

new system

$$\Rightarrow x + 2y - z = 2 \quad (1)$$

$$3y + z = 4 \quad (2)$$

$$-5y + 3z = -2 \quad (3')$$

$$\Rightarrow 3(-5y + 3z) + 5(3y + z) = 3(-2) + 5(4)$$

$$\Rightarrow -15y + 9z + 15y + 5z = -6 + 20$$

$$\Rightarrow 14z = 14$$

$$\Rightarrow z = 1$$

$$\text{From (2): } 3y + 1 = 4 \Rightarrow 3y = 3 \Rightarrow y = 1$$

$$\text{From (1): } x + 2(1) - 1 = 2 \Rightarrow x + 1 = 2 \Rightarrow x = 1$$

$$\boxed{x = 1, y = 1, z = 1}$$

c)

$$2x + y - 4z = -7 \quad (1)$$

$$x - y + z = -2 \quad (2)$$

$$-x + 3y - 2z = 6 \quad (3)$$

$$\Rightarrow x - y + z = -2 \quad (1')$$

$$2x + y - 4z = -7 \quad (2')$$

$$-x + 3y - 2z = 6 \quad (3')$$

compute $(2') - 2(1')$

$$\Rightarrow 2x + y - 4z - 2(x - y + z) = -7 - 2(-2)$$

$$\Rightarrow 2x + y - 4z - 2x + 2y - 2z = -3$$

$$\Rightarrow 3y - 6z = -3 \quad (2'')$$

compute $(3') + (1')$

$$\Rightarrow -x + 3y - 2z + x - y + z = 6 - 2$$

$$\Rightarrow 2y - z = 4$$

New System

$$x - y + z = -2 \quad (1')$$

$$3y - 6z = -3 \quad (2'')$$

$$2y - z = 4 \quad (3'')$$

$$(2'')/3$$

$$\Rightarrow y - 2z = -1 \quad (2''')$$

compute $(3'') - 2(2''')$

$$\Rightarrow 2y - z - 2(y - 2z) = 4 - 2(-1)$$

$$\Rightarrow 3z = 6$$

$$\Rightarrow z = 2$$

$$\text{From } (2''') : y - 2(2) = -1 \Rightarrow y - 4 = -1 \Rightarrow y = 3$$

$$\text{From } (1') : x - 3 + 2 = -2 \Rightarrow x - 1 = -2 \Rightarrow x = -1$$

$$\boxed{x = -1, y = 3, z = 2}$$

2. EX. 2.1.6, Sauer 3

$$\text{Back substitution operations} = n^2 = (5000)^2 = 25,000,000$$

$$\text{Time per operation} = \frac{0.005}{25,000,000} = 2 \times 10^{-10}$$

$$n = 5000$$

$$\text{Elimination operation} = \frac{2n^3}{3} = \frac{2(5000)^3}{3} \approx 8.3333 \times 10^{10} \text{ operations}$$

$$\text{Time} = (8.3333 \times 10^{10}) \times (2 \times 10^{-10})$$

$$\text{Time} \approx 16.6667 \text{ sec}$$

$$\boxed{\text{Time} \approx 17 \text{ sec}}$$

3. EX. 2.1.8, Sauer 3

$$n = 3000$$

$$\frac{2n^3}{3} = \frac{2(3000)^3}{3} = 1.8 \times 10^{10}$$

$$n^2 = (3000)^2 = 9 \times 10^6$$

$$\text{Total operations} = (1.8 \times 10^{10}) + (9 \times 10^6) = 18,009,000,000 \text{ operations}$$

$$\text{Time per operation} = \frac{5}{18,009,000,000} \approx 2.7765 \times 10^{-10}$$

$$\text{Time for one back substitution} = n^2 \times (\text{Time per operation})$$

$$\Rightarrow (9 \times 10^6) \times (2.7765 \times 10^{-10}) \approx 0.0025 \text{ sec}$$

$$\text{Back substitution per second} = \frac{1}{0.0025} \approx \boxed{400}$$

4 Ex. 2.2.2. GZC, saved 3

$$b) \begin{pmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{pmatrix}$$

$$R_2 \rightarrow R_2 - 1 \cdot R_1 \quad (m_{21} = 4/4 = 1)$$

$$R_3 \rightarrow R_3 - \frac{1}{2} \cdot R_1 \quad (m_{31} = 2/4 = \frac{1}{2})$$

$$A = \begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{pmatrix}$$

$$R_3 \rightarrow R_3 - \frac{1}{2} R_2 \quad (m_{32} = \frac{1}{2})$$

$$U = \begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

check $LU = A$

Row 1: $(4, 2, 0)$ ✓

Row 2: $(4, 2 + 2, 2) = (4, 4, 2)$ ✓

Row 3: $(2, 1 + 1, 1 + 2) = (2, 2, 3)$ ✓

Matches A ✓

c)

$$\begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 2 & 1 & 0 \\ 1 & 3 & 4 & 4 \\ 0 & 2 & 1 & -1 \end{pmatrix}$$

$$R_2 \rightarrow R_2 - 0 \cdot R_1 \quad (m_{21} = 0)$$

$$R_3 \rightarrow R_3 - 1 \cdot R_1 \quad (m_{31} = 1)$$

$$R_4 \rightarrow R_4 - 0 \cdot R_1 \quad (m_{41} = 0)$$

$$\begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 2 & 1 & 0 \\ 0 & 4 & 3 & 2 \\ 0 & 2 & 1 & -1 \end{pmatrix}$$

$$R_3 \rightarrow R_3 - 2 \cdot R_2 \quad (m_{32} = 2)$$

$$R_4 \rightarrow R_4 - 1 \cdot R_2 \quad (m_{42} = 1)$$

$$\begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$R_4 \rightarrow R_4 - 0 \cdot R_3 \quad (m_{43} = 0)$$

$$U = \begin{pmatrix} 1 & -1 & 1 & 2 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$$LU = A$$

$$\text{Row 1: } (1, -1, 1, 2) \checkmark$$

$$\text{Row 2: } (0, 2, 1, 0) \checkmark$$

$$\text{Row 3: } (1, 3, 4, 4) \checkmark$$

$$\text{Row 4: } (0, 2, 1, -1) \checkmark$$

matches A $\checkmark$

5. EX. 2.2.4.b Solver 3

$$b) \quad A = \begin{pmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{pmatrix} \quad b = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1/2 & 1/2 & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$Ly = b$$

$$Ux = y$$

$$\Rightarrow Ly = b$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1/2 & 1/2 & 1 \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$y_0 = 2$$

$$y_0 + y_1 = 4$$

$$\frac{1}{2} y_0 + \frac{1}{2} y_1 + \frac{1}{2} y_2 = 6$$

$$y_1 = 4 - y_0 = 4 - 2 = 2$$

$$y_2 = 6 - \frac{1}{2} y_0 - \frac{1}{2} y_1 = 6 - 1 - 1 = 4$$

$$y = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$$

$$V_F = y$$

$$\begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$$

$$4x_0 + 2x_1 = 2$$

$$2x_1 + 2x_2 = 2$$

$$2x_2 = 4$$

$$x_2 = \frac{4}{2} = 2$$

$$2x_1 + 2x_2 = 2 \Rightarrow 2x_1 + 2(2) = 2 \Rightarrow 2x_1 + 4 = 2 \Rightarrow 2x_1 = -2 \Rightarrow x_1 = -1$$

$$4x_0 + 2x_1 = 2 \Rightarrow 4x_0 + 2(-1) = 2 \Rightarrow 4x_0 - 2 = 2 \Rightarrow 4x_0 = 4 \Rightarrow x_0 = 1$$

$$x = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

$$\boxed{x_0 = 1, x_1 = -1, x_2 = 2}$$

checks

$$\text{Row 1: } 4(1) + 2(-1) + 0(2) = 2$$

$$\text{Row 2: } 4(1) + 4(-1) + 2(2) = 4$$

$$\text{Row 3: } 2(1) + 2(-1) + 3(2) = 6$$

6. EX 2.2, 6, save 3

$$\frac{2n^3}{3} = \frac{2(1000)^3}{3} \approx 6.6667 \times 10^8 \text{ operations}$$

cost of 500 solves

$$5 \times 2(1000)^2 = 1 \times 10^9 \text{ operations}$$

total operations

$$(6.6667 \times 10^8) + (1 \times 10^9) = 1.66667 \times 10^9 \text{ operations}$$

$$\text{Time per operation} = \frac{60}{1.6667 \times 10^9} = 3.6 \times 10^{-8} \text{ sec/op}$$

$$\text{Time}_{LU} = (6.6667 \times 10^8) \times (3.6 \times 10^{-8})$$

$$\boxed{\text{Time}_{LU} = 24 \text{ sec}}$$

7. EX. 2.2.8, Save 83

$$\frac{2n^3}{3} + 2n^2 = \frac{2(2000)^3}{3} + 2(2000)^2 \approx 5.333341 \times 10^9$$

Time per operation

$$\frac{0.1}{5.333341 \times 10^9} \approx 1.875 \times 10^{-11}$$

LU factorization cost $n = 8000$

$$\frac{2(8000)^3}{3} \approx 3.41333 \times 10^{11}$$

cost of 100 solves

$$100 \times 2n^2 = 100 \times 2(8000)^2 = 1.28 \times 10^{10}$$

Total operations

$$(3.41333 \times 10^{11}) + (1.28 \times 10^{10}) = 3.54133 \times 10^{11}$$

$$\text{Time} = (3.54133 \times 10^{11}) \times (1.875 \times 10^{-11})$$

$$\text{Time} \approx 6.64 \text{ sec}$$

10. EX. 2.3. 2.62C, Savitzky

$$b) \quad A = \begin{pmatrix} 1 & 2.01 \\ 3 & 6 \end{pmatrix}$$

$$\text{cond}_{\infty}(A) = \|A\|_{\infty} \cdot \|A^{-1}\|_{\infty}$$

$$\|A\|_{\infty} = \max_j \sum_i |a_{ij}|$$

$$\text{Row 1: } |1| + |2.01| = 3.01$$

$$\text{Row 2: } |3| + |6| = 9$$

$$\|A\|_{\infty} = \max(3.01, 9) = 9$$

$$\det(A) = (1)(6) - (2.01)(3) = -0.03$$

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} 6 & -2.01 \\ -3 & 1 \end{pmatrix} = \frac{1}{-0.03} \begin{pmatrix} 6 & -2.01 \\ -3 & 1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} -200 & 67 \\ 100 & -33.33 \end{pmatrix}$$

$$\text{Row 1: } |-200| + |67| = 267$$

$$\text{Row 2: } |100| + |-33.33| = 133.33$$

$$\|A^{-1}\|_{\infty} = \max(267, 133.33) = 267$$

$$\text{cond}_{\infty}(A) = 9 \times 267 = \boxed{2403}$$

$$c) \quad A = \begin{pmatrix} 6 & 3 \\ 4 & 2 \end{pmatrix}$$

$$\det(A) = (6)(2) - (3)(4) = 0$$

$$\text{cond}_\infty(A) = \infty \quad \text{undefined}$$

13. 2.4.2. (2) sauer

$$c) \quad A = \begin{pmatrix} 1 & 2 & -3 \\ 2 & 4 & 2 \\ -1 & 0 & 3 \end{pmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\begin{pmatrix} 2 & 4 & 2 \\ 1 & 2 & -3 \\ -1 & 0 & 3 \end{pmatrix}$$

$$P_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$m_{21} = 1/2, \quad R_2 \rightarrow R_2 - \frac{1}{2} R_1$$

$$m_{31} = -1/2, \quad R_3 \rightarrow R_3 + \frac{1}{2} R_1$$

$$\begin{pmatrix} 2 & 4 & 0 \\ 0 & 0 & -4 \\ 0 & 2 & 4 \end{pmatrix}$$

$$R_2 \longleftrightarrow R_3 \quad \begin{pmatrix} 2 & 4 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & -4 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$U = \begin{pmatrix} 2 & 4 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & -4 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$PA = \begin{pmatrix} 2 & 4 & 2 \\ -1 & 0 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$R_2 \rightarrow R_2 + \frac{1}{2} R_1 \quad m_{21} = -\frac{1}{2}$$

$$R_3 \rightarrow R_3 - \frac{1}{2} R_1 \quad m_{31} = \frac{1}{2}$$

$$m_{32} = 0$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{pmatrix}$$

$$U_1 = (2, 4, 2) \quad U_2 = (0, 2, 4) \quad U_3 = (0, 0, -4)$$

LU Rows:

$$\text{Row 1} \rightarrow 1 \cdot v_1 = (2, 4, 2)$$

$$\text{Row 2} \rightarrow \frac{1}{2} v_1 + v_2 = (-1, 0, 3)$$

$$\text{Row 3} \rightarrow \frac{1}{2} v_1 + 0 \cdot v_2 + v_3 = (1, 2, -3)$$

$$PA = \begin{pmatrix} 2 & 4 & 2 \\ -1 & 0 & 3 \\ 1 & 2 & -3 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 1/2 & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 4 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & -4 \end{pmatrix}$$

d)

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ -2 & 1 & 0 \end{pmatrix}$$

$$R_1 \leftrightarrow R_3 \quad \begin{pmatrix} -2 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 2 \end{pmatrix} \quad P_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$m_{21} = -1/2, \quad R_2 \rightarrow R_2 + \frac{1}{2} R_1$$

$$m_{31} = 0, \quad R_3 \rightarrow R_3 - 0 \cdot R_1$$

$$\begin{pmatrix} -2 & 1 & 0 \\ 0 & 1 & -1/2 \\ 0 & 1 & 0 \end{pmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\begin{pmatrix} -2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1/2 & 2 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$m_{32} = 1/2, \quad R_3 \rightarrow R_3 - \frac{1}{2}R_2$$

$$U = \begin{pmatrix} -2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$PA = \begin{pmatrix} -2 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/2 & 1/2 & 1 \end{pmatrix}$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/2 & 1/2 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/2 & 1/2 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} -2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

14. EX. 2.4.4, solver 3

a)

$$A = \begin{pmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{pmatrix} \quad b = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$P = I$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1/2 & 1/2 & 1 \end{pmatrix} \quad U = \begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

$$Ly = b$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1/2 & 1/2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$y_1 = 2$$

$$y_1 + y_2 = 4 \Rightarrow y_2 = 2$$

$$\frac{1}{2}y_1 + \frac{1}{2}y_2 + y_3 = 6 = 6 - 1 - 1 = 4$$

$$y = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$$

$$U_x = y$$

$$\begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}$$

$$2x_3 = 4 \Rightarrow x_3 = 2$$

$$2x_2 + 2x_3 = 2 \Rightarrow x_2 = -1$$

$$4x_1 + 2x_2 = 2 \Rightarrow x_1 = 1$$

$$\boxed{x_1 = 1, x_2 = -1, x_3 = 2}$$

$$b) \quad A = \begin{pmatrix} -1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} -2 \\ 17 \\ 3 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ -1r_2 & 1 & 0 \\ -1r_2 & 1r_5 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & s_{12} & 1r_2 \\ 0 & 0 & 7r_5 \end{pmatrix}$$

$$P \neq I$$

$$P_b = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} -2 \\ 17 \\ 3 \end{pmatrix} = \begin{pmatrix} 17 \\ 3 \\ -2 \end{pmatrix}$$

$$Ly = Pb$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ -1/2 & 1/5 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 17 \\ 3 \\ -2 \end{pmatrix}$$

$$y_1 = 17$$

$$-\frac{1}{2} y_1 + y_2 = 3 \Rightarrow y_2 = 3 + \frac{1}{2}(17) = \frac{23}{2}$$

$$-\frac{1}{2} y_1 + \frac{1}{5} y_2 + y_3 = -2 \Rightarrow y_3 = -2 + \frac{1}{2}(17) - \frac{1}{5}\left(\frac{23}{2}\right)$$

$$\Rightarrow \frac{21}{5}$$

$$y = \begin{pmatrix} 17 \\ 23/2 \\ 21/5 \end{pmatrix}$$

$$u_x = y$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 5/2 & 1/2 \\ 0 & 0 & 7/5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 17 \\ 23/2 \\ 21/5 \end{pmatrix}$$

$$7/5 x_3 = \frac{21}{5} \Rightarrow x_3 = \frac{21}{5} \cdot \frac{5}{7} = \frac{21}{7} = 3$$

$$5/2 x_2 + \frac{1}{2} x_3 = \frac{23}{2} \Rightarrow x_2 = 4$$

$$2x_1 + x_2 + x_3 = 17 \Rightarrow x_1 = 5$$

$$x_1 = 9, \quad x_2 = 4, \quad x_3 = 3$$

15. EX. 2.7.2.6 $\hookrightarrow$ answer

$$L(x) = F(x_0) + J_F(x_0)(x - x_0)$$

$$F\left(\begin{pmatrix} u \\ v \end{pmatrix}\right) = \begin{pmatrix} u + e^{u-v} \\ 2u + v \end{pmatrix}, \quad x_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$F\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 + e^{1-1} \\ 2(1) + 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\text{Let, } F_1(u, v) = u + e^{u-v} \text{ and } F_2(u, v) = 2u + v$$

$$\Rightarrow \frac{\partial F_1}{\partial u} = 1 + e^{u-v}, \quad \frac{\partial F_1}{\partial v} = -e^{u-v}$$

$$\Rightarrow \frac{\partial F_2}{\partial u} = 2, \quad \frac{\partial F_2}{\partial v} = 1$$

$$\Rightarrow J_F(u, v) = \begin{pmatrix} 1 + e^{u-v} & -e^{u-v} \\ 2 & 1 \end{pmatrix}$$

$$J_F(1, 1) = \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix}$$

$$L(x) = f(x_0) + Jf(x_0)(x - x_0)$$

$$L\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} u-1 \\ v-1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} u-1 \\ v-1 \end{pmatrix} = \begin{pmatrix} 2(u-1) & -(v-1) \\ 2(u-1) & +(v-1) \end{pmatrix} = \begin{pmatrix} 2u - u - 1 \\ 2u + v - 3 \end{pmatrix}$$

$$\Rightarrow L\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2u - u - 1 \\ 2u + v - 3 \end{pmatrix} = \begin{pmatrix} 2u - u + 1 \\ 2u + v \end{pmatrix}$$

$$L = \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} u-1 \\ v-1 \end{pmatrix} = \begin{pmatrix} 2u - u + 1 \\ 2u + v \end{pmatrix}$$